

Cambridge International AS & A Level

CHEMISTRY**9701/33**

Paper 3 Advanced Practical Skills 1

February/March 2026**MARK SCHEME**Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the February/March 2026 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **13** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	Correct point or mark awarded
	Incorrect point or mark not awarded
	Information missing or insufficient for credit
	Benefit of the doubt given
	Contradiction in response otherwise markworthy, mark not given
	Error in number of decimal places
	Error carried forward applied
	Incorrect or insufficient point ignored while marking the rest of the response
	Rounding error
	Repeat error
 or /	Blank page or part of script seen

Annotation	Meaning
SF	Error in number of significant figures
TE	Transcription error

Mark Scheme abbreviations

;	separates marking points
/	alternative responses for the same marking point
R	reject the response
A	accept the response
I	ignore the response
ecf	error carried forward
AVP	any valid point
ora	or reverse argument
AW	alternative wording
<u>underline</u>	actual word given must be used by candidate (grammatical variants excepted)
()	the word / phrase in brackets is not required but sets the context
max	indicates the maximum number of marks that can be given
M1 etc.	marking point
owtte	or words to that effect

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Question	Answer	Marks									
1(a)	I 6 unambiguous headings for mass, with correctly displayed units, recorded in Results space, for 1(a) and 1(c) : (Mass of) container + FA 1 / g (Mass of) container (empty or with residual FA 1) / g - Mass of FA 1 (used) / g (Mass of) container + FA 3 / g (Mass of) container (empty or with residual FA 3) / g - Mass of FA 3 (used) / g	1									
	II 6 unambiguous headings for temperature, with correctly displayed units, recorded in Results space, for 1(a) and 1(c) : - Initial / starting temperature / °C - Minimum / maximum / final temperature / °C - Temperature change (or decrease) / °C	1									
	III Precision of readings in (a) and (c) - All four balance readings recorded to same number of decimal places (2 or 3). AND - All four thermometer readings recorded to .0 or .5 °C.	1									
	IV Accuracy mark Calculate the temperature change, ΔT, for Supervisor and each candidate. <table><tr><td>Supervisor temperature change, ΔT/°C</td><td>Award mark IV if δ, the difference between the supervisor ΔT and candidate ΔT, is:</td></tr><tr><td>ΔT ≤ 5.0</td><td>± 0.5 °C</td></tr><tr><td>5.0 < ΔT ≤ 10.0</td><td>± 1.0 °C</td></tr><tr><td>10.0 < ΔT ≤ 15.0</td><td>± 1.5 °C</td></tr><tr><td>15.0 < ΔT</td><td>± 2.0 °C</td></tr></table>	Supervisor temperature change, ΔT/°C	Award mark IV if δ, the difference between the supervisor ΔT and candidate ΔT, is:	ΔT ≤ 5.0	± 0.5 °C	5.0 < ΔT ≤ 10.0	± 1.0 °C	10.0 < ΔT ≤ 15.0	± 1.5 °C	15.0 < ΔT	± 2.0 °C
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10.0 < ΔT ≤ 15.0	± 1.5 °C										
15.0 < ΔT	± 2.0 °C										

Question	Answer	Marks									
1(b)(i)	Energy change = $25 \times 4.18 \times \Delta T$ AND answer correctly rounded to 2–4 sf	1									
1(b)(ii)	$n(\text{carbonate}) = \frac{\text{mass used}}{286} \text{ mol}$ AND answer correctly rounded to 2–4 sf	1									
1(b)(iii)	Correct use of data and correct sign for the data collected $\Delta H_1 = \frac{(b)(i)}{(b)(ii)} \times \frac{1}{1000} \text{ kJ mol}^{-1}$ AND answer correctly rounded to 2–4 sf	1									
1(c)(i)	I Correct subtractions in (a) and (c) - Mass of FA 1 and FA 3 used AND - both temperature changes	1									
	II Accuracy mark Calculate the temperature change, ΔT , for Supervisor and each candidate. <table><tr><td>Supervisor temperature change, $\Delta T / ^\circ\text{C}$</td><td>Award mark IV if δ, the difference between the supervisor and candidate ΔTs, is:</td></tr><tr><td>$\Delta T \leq 5.0$</td><td>$\pm 0.5 ^\circ\text{C}$</td></tr><tr><td>$5.0 < \Delta T \leq 10.0$</td><td>$\pm 1.0 ^\circ\text{C}$</td></tr><tr><td>$10.0 < \Delta T \leq 15.0$</td><td>$\pm 1.5 ^\circ\text{C}$</td></tr><tr><td>$15.0 < \Delta T$</td><td>$\pm 2.0 ^\circ\text{C}$</td></tr></table>	Supervisor temperature change, $\Delta T / ^\circ\text{C}$	Award mark IV if δ , the difference between the supervisor and candidate ΔT s, is:	$\Delta T \leq 5.0$	$\pm 0.5 ^\circ\text{C}$	$5.0 < \Delta T \leq 10.0$	$\pm 1.0 ^\circ\text{C}$	$10.0 < \Delta T \leq 15.0$	$\pm 1.5 ^\circ\text{C}$	$15.0 < \Delta T$	$\pm 2.0 ^\circ\text{C}$
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$10.0 < \Delta T \leq 15.0$	$\pm 1.5 ^\circ\text{C}$										
$15.0 < \Delta T$	$\pm 2.0 ^\circ\text{C}$										

Question	Answer	Marks
1(c)(ii)	M1: correct expressions: At least 2 correct expressions shown: - energy / J = $30 \times 4.18 \times \Delta T$ - amount / mol = $\frac{\text{mass used}}{106}$ - enthalpy = $\frac{\text{energy}}{\text{amount} (\times 1000)}$ M2: correct use of data negative sign AND correct numerical answer to 2–4 sf	2
1(d)	Correct use of candidate's own data $\Delta H_r = \text{(c)(ii)} - \text{(b)(iii)}$ AND answer to at least 2 sf	1
1(e)	EITHER Student is not correct AND explanation that acid <u>is still</u> in excess AND temperature rise will be the same OR Student is not correct AND explanation that reaction will be faster OR concentration of $\text{H}^+(\text{aq})$ ions is higher AND the quantity of frothing / spitting / acid spray will be greater OR Student is correct plus explanation that reaction will be faster or concentration of $\text{H}^+(\text{aq})$ ions is higher AND less (time for) heat loss (to surroundings).	1

Question	Answer	Marks
2(a)	I The following data are recorded: - two burette readings AND titre for the rough titration - initial and final burette readings for two (or more) accurate titrations	1
	II Titre values recorded for accurate titrations, AND correct headings and units in the accurate titration table - initial / start AND (burette) reading / volume - final / end AND (burette) reading / volume - titre OR volume used / added or FA 5 used / added - unit: / cm³ OR (cm³) OR in cm³ (for each heading) OR cm³ unit given for each volume recorded.	1
	III All accurate burette readings are recorded to the nearest 0.05 cm ³ .	1
	IV The final accurate titre recorded is within 0.10 cm ³ of any other accurate titre.	1
	Accuracy (Q) marks Round burette readings to the nearest 0.05 cm ³ . Check and correct titre subtractions where necessary. Examiner selects the best mean titre, using the following hierarchy: - two (or more) accurate identical titres (ignoring any that are labelled 'rough'), <i>then</i> - two (or more) accurate titres within 0.05 cm³, <i>then</i> - two(or more) accurate titres within 0.10 cm³, <i>etc.</i> Calculate the difference (δ) between the candidate's mean titre and the supervisor's mean titre	
	Award accuracy Q marks as follows: V Award if $\delta \leq 0.50 \text{ cm}^3$ VI Award if $\delta \leq 0.30 \text{ cm}^3$ VII Award if $\delta \leq 0.20 \text{ cm}^3$	3
2(b)	Correctly calculates mean titre to 2 d.p. - Candidate must take the average of two (or more) titres that are within a total spread of not more than 0.20 cm³. - Working / explanation must be shown OR ticks must be put next to the two (or more) accurate readings selected. - The mean should be quoted to 2 d.p. and be rounded to nearest 0.01 cm³.	1

Question	Answer	Marks
2(c)(i)	All answers given to parts (c)(ii), (c)(iii), (c)(iv) and (c)(v) recorded to 3 or 4 sf.	1
2(c)(ii)	Correct calculation Amount HC/ used = $0.110 \times {}^{(b)}/_{1000}$ mol	1
2(c)(iii)	Correct use Amount carbonate = $0.5 \times (c)(ii) \times 40 \text{ mol dm}^{-3}$	1
2(c)(iv)	Correct use Average $M_r = {}^{5.85}/_{(c)(iii)}$	1
2(c)(v)	Correct use $106 p + 138.2 (1 - p) = (c)(iv)$ (where $p = \% \text{ of } Na_2CO_3 / 100$)	1

Question	Answer	Marks
FA 7 and FA 8 are $(NH_4)_2CO_3$; FA 9 is NaOH		
3(a)(i)	2 • = 1 mark • White solid / powder or white / colourless crystals (at start) • condensation / steam / steamy fumes / water droplets / moisture • (condensation forms on) gentle heating / soon after heating began • No residue (at end) / solid evaporates / solid sublimates / solid reduces in quantity • (Gas / ammonia) turns (red) litmus blue	2
3(a)(ii)	M1: Fizzing / bubbling / effervescence AND EITHER rapid / strong / violent / vigorous (reaction) OR colourless solution formed / colourless solution remains / solid dissolves M2: Gas / CO_2 gives <u>white precipitate</u> with lime water	2

Question	Answer			Marks
3(b)(i)	2 • = 1 mark			7
	test	FA 8	FA 9	
	Test 1 Ba ²⁺ (aq)	White <u>ppt</u>	(Slight) white <u>ppt</u> AND (ppt) soluble / dissolves / forms colourless solution (in acid) OR No change / no reaction (<i>owtte</i>) in both boxes	
	+ acid	(ppt) soluble / dissolves OR colourless <u>solution formed</u> - fizzing - gas / CO₂ gives white ppt with limewater (credit once only in 3(b)(i))		
	Test 2 AgNO ₃	- White / off-white <u>ppt</u> (ppt) soluble / dissolves/ <u>forms</u> colourless solution	brown / grey / black ppt	
	+ HNO ₃	- Fizzing - Gas / CO₂ gives white ppt with limewater (credit once only 3(b)(i))	(ppt) soluble / dissolves / <u>forms</u> colourless solution	
	Test 3 + FA 9	no change / no ppt / no reaction (in cold) / remains colourless (solution)		
	warm	<u>gas / ammonia</u> turns litmus to blue		
	Test 4 ZnSO ₄	- white ppt (ppt) insoluble in excess	- white ppt (ppt) soluble in excess	
	Test 5 litmus		red (litmus) goes blue AND blue stays blue / goes darker blue	

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Question	Answer	Marks
3(b)(ii)	2 or 3 correct = 1 mark. All 4 correct = 2 marks. FA 8: ions are NH_4^+ and CO_3^{2-} FA 9: ions are 'unknown' and OH^-	2
3(b)(iii)	$(\text{NH}_4)_2\text{CO}_3(\text{s}) \rightarrow 2\text{NH}_3(\text{g}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$	1